

# MUHAMMAD HUSNAIN MUBARIK

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## RESEARCH INTERESTS

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Computer Architecture, Hardware Acceleration of Machine Learning Applications, Hardware for Graphics (Conventional, Neural and Generative), Hardware Acceleration of Generative AI, Printed Computer Systems, Emerging Technologies and Applications.

## EDUCATION

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**University of Illinois at Urbana Champaign (UIUC), USA.** *Jan 2019 - May 2024*  
PhD - Electrical and Computer Engineering (ECE)  
Advisor: [Prof. Rakesh Kumar](#)

**National University of Sciences and Technology (NUST), Pakistan.** *Sep 2013 - July 2017*  
Bachelors of Science in Electrical Engineering.  
CGPA: 3.95/4.00, **summa cum laude**.

## PUBLICATION SUMMARY AND IMPACT

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- 6× (ISCA, MICRO), 1× (DATE)
- 1× ISCA Retrospective (1996-2020) Selection
- 2× IEEE Top Picks Honorable Mention
- Dan Vivoli Endowed Fellowship
- Served as a program committee (PC) member at HPCA, ISCA, ASPLOS, MLSys, and DAC.
- Emerging Vision and Graphics Systems and Architectures (EVGA), workshop organization, co-located with ISCA 2024.

## RESEARCH AND WORK EXPERIENCE

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### Advanced Rendering Research (ARR), AMD, USA.

- Senior Member of Technical Staff (SMTS) Software Development Eng. *June 2024 - **Present***  
Full-time, Manager: Bruno Stefanizzi
- From Bit-Position Sensitivity to Unequal Error Protection for DNN Inference Memory (**MICRO-59th, 2026**)

### Graduate Research Intern, Intel, USA.

- **Emerging Visual-AI Systems Research Lab** *February 2024 - May 2024*  
Part-time (20 hrs), Managers: Nilesh Jain, Ravishankar Iyer.
- **Emerging Visual-AI Systems Research Lab** *September 2023 - February 2024*  
Part-time (10 hrs), Managers: Nilesh Jain, Ravishankar Iyer.
- **Graphics Research Organization (GRO)** *May 2023 - August 2023*  
Full-time, Managers: Tobias Zirr, Anton Sochenov, Anton Kaplanyan.
- **Next-Gen Architecture (NGA)** *September 2022 - May 2023*  
Part-time, Manager: Maxim Kazakov.

- **Graphics Research Organization (GRO)** *June 2022 - August 2022*  
Full-time, Managers: Rama Hariharan, Anton Kaplanyan.
- **Cloud Systems Research Lab (CSR)** *December 2021 - May 2022*  
Part-time, Managers: Nilesch Jain, Ravishankar Iyer.
- **Heterogeneous Platforms Lab (HPL)** *May 2021 - August 2021*  
Full-time, Manager: Tanay Karnik.

## **Coordinated Science Laboratory (CSL), UIUC, USA**

*January 2019 - May 2024*

Research Assistant

Advisor: Prof. Rakesh Kumar

- Hardware Acceleration of Latent Diffusion Models (Undersubmission)
- ML accelerators for accelerating gaming and vision applications (Ongoing)
- Hardware Acceleration of Neural Graphics (**ISCA-50th, 2023**)
- Understanding Interactions Between Chip Architecture and Uncertainties in Semiconductor Supply and Demand (Under Submission - Preprint available on arXiv.)
- Exploiting Short Application Lifetimes for Low Cost Hardware Encryption in Flexible Electronics (**DATE 2023**)
- Rethinking Programmable Earable Processors (**ISCA-49th, 2022**)
- Model-specific Design of Deeply-Embedded Tiny Neural Network Accelerators (Ongoing).
- Printed Machine Learning Classifiers (**MICRO-53rd, 2020**)
- Printed Microprocessors (**ISCA-47th, 2020**)

## **PUBLICATIONS**

- 1 **M. Mubarik**, K. Mohan Kumar, P. A. Pena, K. Varadarajan, K. Tyagi, "From Bit-Position Sensitivity to Unequal Error Protection for DNN Inference Memory," **MICRO-59th, 2026**.
- 2 N. Bleier, **M. Mubarik**, G. Swenson, R. Kumar "Space Microdatacenters" **MICRO-56th, 2023**, **IEEE Micro Top Picks - Honorable Mention 2024**.
- 3 **M. Mubarik**, R. Kanungo, T. Zirr, R. Kumar "Hardware Acceleration of Neural Graphics" **ISCA-50th, 2023**.
- 4 N. Bleier, **M. Mubarik**, S. Balaji, F. Rodriguez, A. Sou, S. White and R. Kumar, "Exploiting Short Application Lifetimes for Low Cost Hardware Encryption in Flexible Electronics," **DATE 2023**.
- 5 N. Bleier, **M. Mubarik**, S. Chakraborty, S. Kishore, R. Kumar, "Rethinking Programmable Earable Processors," **ISCA-49th, 2022**.
- 6 **M. Mubarik**, D. Weller, N. Bleier, M. Tomei, J. Aghassi-Hagmann, M. Tahoori, R. Kumar, "Printed Machine Learning Classifiers," **MICRO-53rd 2020**, **IEEE Micro Top Picks - Honorable Mention 2021**.
- 7 N. Bleier\*, **M. Mubarik\***, F. Rasheed\*, J. Aghassi-Hagmann, M. Tahoori, R. Kumar, "Printed Microprocessors," **ISCA-47th, 2020**. [\* Co-First Authors, Listed in Alphabetical Order] **Selected for the retrospective of the years 1996 through 2020 on the 50th anniversary of ISCA**.

- 8 R. Kanungo, S. Siva, N. Bleier, **M. Mubarik**, L. Varshney, R. Kumar "Understanding Interactions Between Chip Architecture and Uncertainties in Semiconductor Supply and Demand" - Under Submission, preprint arXiv:2305.11059, 2023.

## TALKS

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- Delivered a crash course on diffusion models at AMD, December 2024.
- Delivered a guest lecture on "Neural Rendering" at AMD, June 2024.
- Delivered a guest lecture on "Tensor Processing Units" at Intel Labs, April 2024.
- Delivered a guest lecture on "Accelerators/Specialization/Emerging Architectures" in ECE 411, Spring 2024.
- Delivered a guest lecture on "GPU Architecture and Programming" in ECE 411, Spring 2024.
- Gave a talk titled "Hardware Acceleration Opportunities in Generative AI Based Graphics" at Intel [2024].
- Presented "Hardware Acceleration of Neural Graphics" paper at ISCA-50th, 2023.
- Gave a talk on "Neural Graphics: An Architecture's Perspective" at ECE498SJP: Accelerator Architectures class at UIUC [2023].
- Gave a talk titled "Hardware Acceleration Opportunities in Neural Radiance Fields" at Intel [2022].
- Presented "Printed Machine Learning Classifiers" paper at MICRO-53rd, 2020.
- Gave a talk on "Hardware for Deep Learning" at ECE511: Advanced Computer Architecture class at UIUC [2021].

## SELECT ACADEMIC ACHIEVEMENTS AND AWARDS

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- 🏆 **Dan Vivoli Endowed Fellowship (NVIDIA) for the 2024-2025 academic year.**
- 🏆 **IEEE Micro Top Picks - Honorable Mention, 2024.**
  - Our paper "Space Microdatacenters" (MICRO-56th, 2023) ranks among the Top 24 papers published in computer architecture venues in 2023.
- 🏆 **ISCA-50th 25 year retrospective (1996-2020), 2023.**
  - Selected for the retrospective of the years 1996 through 2020 on the 50th anniversary of ISCA for our paper "Printed Microprocessors" (ISCA-47th, 2020).
- 🏆 **Machine Learning and Systems Rising Star, 2023.**
- 🏆 **ECE-UIUC travel grant for attending ISCA-2023.**
- 🏆 **IEEE Micro Top Picks - Honorable Mention, 2021.**
  - Our paper "Printed Machine Learning Classifiers" (MICRO-53rd, 2020) ranks among the Top 24 papers published in computer architecture venues in 2020.
- 🏆 **Prime Minister's Silver Medal, NUST-CEME, 2018.**
  - 2nd highest CGPA (3.95/4.00) in the Electrical Engineering Department at NUST-CEME.
- 🏆 **Travel award for EECamp at KAIST, South Korea, 2018.**
- 🏆 **NUST GPA scholarships 2013-2017.**
- 🏆 **High Achievers Award, NUST-CEME, 2015.**

🏆 Golden Star Award, NUST-CEME, 2015.

## TECHNICAL SKILLS

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- Programming languages: C++, Python, Verilog, SystemVerilog, HIP, CUDA, GLSL, Scala, SystemC, Tcl, Bash, Makefile
- Software: PyTorch, TensorFlow, TensorRT, Onnx-Runtime, Caffe, Keras, Synopsys-VCS, MATLAB, LaTeX, Vim
- Frameworks: CUDA, Nvidia Nsight (Compute/Systems/Graphics), Vulkan, OpenGL, CHISEL, gem5.

## PROFESSIONAL SERVICES AND ACTIVITIES

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- Conference Roles:
  - Program Committee (PC) Member:
    - \* International Symposium on Computer Architecture (ISCA) – 2025, 2026.
    - \* Design Automation Conference (DAC) – 2025, 2026.
    - \* Conference on Machine Learning and Systems (MLSys) – 2025, 2026.
    - \* Architectural Support for Programming Languages and Operating Systems (ASPLOS) – 2026.
    - \* International Symposium on High-Performance Computer Architecture (HPCA) – 2025.
    - \* Emerging Vision and Graphics Systems and Architectures (EVGA), co-located with ISCA – 2024.
  - Artifact Evaluation Committee (AEC) Member: ASPLOS – 2025.
  - Reviewer: ACM Transactions on Architecture and Code Optimization (TACO).
- Workshop Organization:
  - ML and Systems Rising Stars, 2024 - Present.
  - Emerging Vision and Graphics Systems and Architectures (EVGA), co-located with the 51st International Symposium on Computer Architecture (ISCA), 2024.

## TEACHING AND MENTORING EXPERIENCE

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- Mentored student projects in ECE 508 (Manycore Parallel Algorithms, Spring 2023) at UIUC.
- Frank Cai 4th-year Undergraduate Student  
Project titled "Hardware Acceleration Opportunities for Speculative Decoding to Accelerate LLMs." Secured admission at UIUC's Graduate College.
- Ramakrishna Kanungo 2nd year Masters Student  
Hardware Accelerator for Neural Radiance Field Inference. Secured industry job in load-store unit design.
- Abigail Wezelis 2nd year Masters Student  
Design space exploration of data specific architectures for accelerating DNNs. Secured industry job in system design.
- Srijan Chakraborty 2nd year Masters Student  
Synthesis and performance/area/power analysis of very large RTL designs.

## SELECT COURSES

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- 📖 ECE498SJP: Accelerator Architectures (Sanjay Patel) - Spring 2023.
- 📖 CS534: Advanced Topics in Computer Architecture (Sarita Adve) - Spring 2022.
- 📖 ECE598JH: Advanced Memory and Storage Systems (Jian Huang) - Fall 2021.
- 📖 CS598DHK: 3D Vision (Derek Hoiem) - Fall 2021.
- 📖 CS498ME Architecture for Mobile and Edge Computing (Saugata Ghose) - Spring 2021.
- 📖 CS598LCE Languages and Compilers for Edge Computing (Vikram S. Adve) - Spring 2021.
- 📖 ECE598NSG: Deep Learning in Hardware (Naresh Shanbhag) - Fall 2020.
- 📖 ECE524/CS563: Advanced Computer Security (Adam Bates) - Spring 2020.
- 📖 CS598SVA: App-Cust Heterogeneous Systems (Sarita Adve) - Spring 2020.
- 📖 CS598JT: Energy Efficient Computer Architecture (Josep Torrellas) - Fall 2019.
- 📖 ECE511: Advanced Computer Architecture (Rakesh Kumar) - Spring 2019.
- 📖 CS533: Parallel Computer Architecture (Josep Torrellas) - Spring 2019.